NINE EDUCATION IIT ACADEMY

CALORIMETRY — PRACTICE ASSIGNMENT

Class 11 Physics · JEE Main Focus · 60 Questions

Specific Heat & Calorimetry · Latent Heat · Mixtures & the Heat Budget Method

FORMULA SHEET Use these formulas and standard values throughout

1. Heat with no change of phase

$$Q = m\,c\,\Delta\theta \qquad (\Delta\theta = \theta_{\text{higher}} - \theta_{\text{lower}},\ \text{always positive})$$

Heat capacity of an object: $C = mc$ — a property of the object, not just the material.

Water equivalent W: the calorimeter behaves as W extra grams of water. Add W to the water's mass in every calculation.

2. Principle of calorimetry (no phase change)

Heat lost by hot bodies = Heat gained by cold bodies (both sides positive).

3. Heat with change of phase (latent heat)

$$Q = m\,L \qquad (\text{no } \Delta\theta \text{ term — temperature is fixed during a phase change})$$

L_f = latent heat of fusion (solid ↔ liquid, at $0°$C)

L_v = latent heat of vaporisation (liquid ↔ gas, at $100°$C)

4. The heat budget method (mixtures)

Before mixing anything, work out where the temperature will settle.

1. Pick the landmark temperature the answer might get stuck at — usually $0°$C or $100°$C.
2. **AVAILABLE** = heat the hot side gives up coming down to that landmark.
3. **REQUIRED** = heat the cold side needs to reach it, including any phase change.
4. Compare the two:
   - If AVAILABLE < REQUIRED, the hot side runs out first. The mixture settles *at* the landmark — find how much of each phase is present.
   - If AVAILABLE > REQUIRED, there is heat to spare. The mixture moves *past* the landmark — write heat lost = heat gained and solve for the final temperature.

STANDARD VALUES (used unless a question states otherwise)

QUANTITY	cal/g-°C or cal/g	J/kg-K or J/kg
c (water)	1	4186 (≈ 4200)
c (ice)	0.5	2100
c (steam)	0.46	1920
L_f (fusion)	80	3.36×10^5
L_v (vaporisation)	540	2.26×10^6
1 cal $= 4.186$ J (use 4.2 J unless told otherwise)		

SECTION A — Specific Heat & Calorimetry Q1–12

1. 150 g of water is heated from $30°$C to $80°$C. How much heat does it absorb?
2. 2400 cal of heat is supplied to 80 g of water initially at $20°$C. Find its final temperature.
3. A 250 g metal piece absorbs 1050 cal and its temperature rises from $25°$C to $45°$C. Find its specific heat and identify the metal from your table.
4. Two blocks of the same material have masses 100 g and 400 g. If both are given the same amount of heat, find the ratio of their temperature rises.
5. A calorimeter has a water equivalent of 25 g. Find how much heat the empty calorimeter absorbs as its temperature rises from $20°$C to $50°$C.
6. 300 g of water at $70°$C is mixed with 200 g of water at $20°$C. Find the temperature of the mixture.
7. Equal masses of two liquids at $30°$C and $70°$C are mixed and the result is $55°$C. Find the ratio of their specific heats.
8. Equal masses of three liquids P, Q, R are at $10°$C, $20°$C and $35°$C. Mixing P and Q gives $14°$C; mixing Q and R gives $25°$C. Find the temperature when P and R are mixed.
9. A 60 g copper piece at temperature T is dropped into a 90 g copper calorimeter containing 150 g water at $25°$C. Final temperature is $35°$C. Take $c_{\text{copper}} = 0.1$ cal/g-°C. Find T.
10. A 40 g metal ball at $200°$C is dropped into 100 g water at $20°$C (calorimeter negligible). Final temperature is $28°$C. Find the metal's specific heat.
11. A 2 kg copper block and a 2 kg lead block each receive 1000 cal. Find the ratio of their temperature rises ($c_{\text{copper}} = 0.09$, $c_{\text{lead}} = 0.03$ cal/g-°C).
12. A 200 g copper calorimeter ($c = 0.1$ cal/g-°C) holds 300 g of water at $15°$C. A 100 g piece of metal at $100°$C is dropped in and the final temperature is $20°$C. Find the specific heat of the metal.

SECTION B — Latent Heat Q13–29

13. Find the heat needed to completely melt 60 g of ice at $0°$C.
14. Find the heat released when 15 g of steam at $100°$C condenses completely at $100°$C.
15. 4000 cal melts a sample of ice at $0°$C completely. Find the mass of ice.
16. 6600 cal melts 30 g of a substance completely at its melting point. Find its latent heat of fusion.
17. Find the heat to raise 25 g of ice from $-8°$C to water at $0°$C.
18. Find the heat to convert 12 g of water at $90°$C completely into steam at $100°$C.
19. Find the total heat to convert 5 g of ice at $-20°$C completely into steam at $100°$C.
20. Repeat in SI: convert 2 g of ice at $-15°$C into steam at $100°$C. Take $c_{\text{ice}} = 2100$, $c_{\text{water}} = 4200$, $L_f = 3.36 \times 10^5$, $L_v = 2.26 \times 10^6$ (all J/kg or J/kg-K). Answer in joules.
21. Find the total heat to convert 8 g of ice at $0°$C into steam at $120°$C ($c_{\text{steam}} = 0.46$ cal/g-°C).
22. Find the total heat to raise 10 g of ice from $-10°$C to water at $60°$C.
23. Find the ratio of heat needed to convert 1 g of water at $100°$C to steam, versus the heat needed to melt 1 g of ice at $0°$C.
24. 2 g of steam at $100°$C and 2 g of boiling water at $100°$C both land on skin. How many extra calories does the steam deliver, purely from condensing?
25. An 18 g ice cube at $-5°$C is converted completely to water at $20°$C. Find the total heat required.
26. Find the heat needed to convert 20 g of ice at $-10°$C into water at $0°$C.
27. Find the heat released when 30 g of steam at $120°$C is cooled to water at $100°$C ($c_{\text{steam}} = 0.46$ cal/g-

°C).

28. How much heat is required to convert 50 g of water at $20°C$ completely into steam at $100°C$?
29. 6400 cal is supplied to 100 g of ice at $0°C$. Find the mass of ice that melts and the mass of ice left over.

SECTION C — Mixtures & the Heat Budget Method Q30–51

30. 10 g of ice at $0°C$ is mixed with 90 g of water at $30°C$. Find the final temperature.
31. 30 g of ice at $0°C$ is mixed with 32 g of water at $20°C$. Find the final temperature and the masses of ice and water present.
32. 25 g of ice at $0°C$ is mixed with 16 g of water at $50°C$. Find the final temperature and the final masses.
33. 200 g of ice at $-40°C$ is mixed with 10 g of water at $5°C$. Find the final temperature.
34. 5 g of steam at $100°C$ is mixed with 45 g of water at $20°C$. Find the final temperature.
35. 5 g of steam at $100°C$ is mixed with 30 g of water at $46°C$. Find the final temperature and masses of water and steam.
36. 4 g of ice at $0°C$ is mixed with 2 g of steam at $100°C$. Find the final temperature and the masses of water and steam present.
37. A calorimeter of water equivalent 10 g contains 50 g of water at $20°C$. Steam at $100°C$ is passed in until the mixture reaches $40°C$. Find the mass of steam condensed.
38. 2550 cal is supplied to 60 g of ice at $-5°C$. Find the final temperature and the masses of ice and water present.
39. 1275 cal is supplied to 15 g of ice at $-10°C$. Find the final state and confirm no heat is left over after the ice just reaches $0°C$ water.
40. 40 g of steam at $100°C$ is mixed with 300 g of ice at $-10°C$. Find the final temperature.
41. 25 g of steam at $100°C$ is mixed with 400 g of ice at $-20°C$. Find the final temperature and the masses of ice and water present.
42. What mass of steam at $100°C$ must be passed into 250 g of water at $15°C$ to raise it to $80°C$?
43. 40 g of ice at $0°C$ is placed in contact with 40 g of water at temperature θ. Find θ such that the water exactly melts all the ice, settling at precisely $0°C$.
44. What is the maximum mass of ice at $0°C$ that 1 kg of water at $50°C$ can completely melt?
45. 60 g of ice at $0°C$ is mixed with water at $45°C$. If exactly 27 g of ice melts before equilibrium at $0°C$, find the mass of water used.
46. 10 g of ice at $0°C$ is mixed with 16 g of water at $15°C$. Find the final temperature and the masses of ice and water present.
47. 4 g of ice at $0°C$ is dropped into 196 g of water at $20°C$. Find the final temperature.
48. A calorimeter of negligible heat capacity holds 100 g of ice at $0°C$. Find the minimum mass of steam at $100°C$ needed to just completely melt all the ice, with the final mixture at exactly $0°C$.
49. 20 g of ice at $0°C$ is mixed with 100 g of water at $40°C$. Find the final temperature.
50. 50 g of ice at $0°C$ is mixed with 20 g of water at $60°C$. Find the final temperature and the masses of ice and water present.
51. A calorimeter of water equivalent 19 g contains 100 g of water at $20°C$. Steam at $100°C$ is passed in until the mixture reaches $45°C$. Find the mass of steam condensed.

JEE MAIN PYQ CORNER Q52–60

Real JEE Main questions, cited by year and shift. Weighted toward Sections B and C, which carry the highest PYQ frequency in this chapter.

52. A copper ball of mass 100 g is at temperature T. It is dropped into a copper calorimeter of mass 100 g containing 170 g of water at room temperature (30°C). The final temperature is 75°C. Take $c_{\text{copper}} = 0.1$ cal/g-°C. Find T. [JEE Main 2017, 02 Apr]

53. Equal masses of three liquids X, Y, Z are at 10°C, 20°C, 30°C. Mixing X and Y gives 16°C; mixing Y and Z gives 26°C. Find the temperature when X and Z are mixed. [JEE Main 2021, 26 Aug Shift 2]

54. A steam engine intakes 50 g of steam at 100°C per minute and cools it down to 20°C. Take $L_v = 540$ cal/g. Find the heat rejected per minute. [JEE Main 2022, 25 June Shift 1]

55. 1 g of ice at -10°C is transformed to vapour at 110°C by applying heat. Take $c_{\text{ice}} = 2100$, $c_{\text{water}} = 4180$, $c_{\text{steam}} = 1920$ (all J/kg-K), $L_f = 3.35 \times 10^5$ J/kg, $L_v = 2.25 \times 10^6$ J/kg. Find the total heat required. [JEE Main 2025, 22 Jan Shift 1]

56. M grams of steam at 100°C is mixed with 200 g of ice at its melting point in an insulated container, producing liquid water at 40°C. Find M. [JEE Main 2020, 07 Jan Shift 2]

57. Take $c_{\text{water}} = 4200$ J/kg-K, $L_f(\text{ice}) = 3.4 \times 10^5$ J/kg. 100 g of ice at 0°C is placed in 200 g of water at 25°C. Find the mass of ice that melts as the water's temperature reaches 0°C. [JEE Main 2020, 04 Sep Shift 1]

58. A calorimeter of water equivalent 20 g contains 180 g of water at 25°C. m grams of steam at 100°C is mixed in until the mixture reaches 31°C. Take $L_v = 540$ cal/g. Find m. [JEE Main 2020, 03 Sep Shift 2]

59. 184 kJ of heat is given to 600 g of ice at -12°C. Take $c_{\text{ice}} = 2222.3$ J/kg-°C, $L_f = 336$ kJ/kg. Find the final state and the ratio of ice to water. [JEE Main 2023, 29 Jan Evening]

60. 10 kg of ice at -10°C is added to 100 kg of water at 25°C, with no heat exchange to surroundings. Find the decrease in the water's temperature. [JEE Main 2026, 28 Jan Morning]

ANSWER KEY Q1–60

Q#	ANSWER
1	7500 cal
2	$50°C$
3	0.21 cal/g-°C (Aluminium)
4	$4:1$
5	750 cal
6	$50°C$
7	$3:5$
8	$16.25°C$
9	$300°C$
10	≈ 0.116 cal/g-°C
11	$3:1$
12	0.2 cal/g-°C
13	4800 cal
14	8100 cal
15	50 g
16	220 cal/g
17	2100 cal
18	6600 cal
19	3650 cal
20	6095 J
21	5833.6 cal
22	1450 cal
23	$6.75:1$
24	1080 cal
25	1845 cal
26	1700 cal
27	16476 cal
28	31000 cal
29	80 g melts; 20 g ice left
30	$19°C$

Q#	ANSWER
31	$0°C$; ice 22 g, water 40 g
32	$0°C$; ice 15 g, water 26 g
33	$-30°C$
34	$82°C$
35	$100°C$; water 33 g, steam 2 g
36	$100°C$; water 5.33 g, steam 0.67 g
37	2 g
38	$0°C$; ice 30 g, water 30 g
39	Exactly $0°C$, all water, nothing left over
40	$\approx 0.29°C$
41	$0°C$; ice 250 g, water 175 g
42	≈ 29.02 g
43	$80°C$
44	625 g
45	48 g
46	$0°C$; ice 7 g, water 19 g
47	$18°C$
48	12.5 g
49	$20°C$
50	$0°C$; ice 35 g, water 35 g
51	5 g
52	$885°C$
53	$\approx 23.85°C$
54	31×10^3 cal
55	3043 J
56	40 g
57	≈ 61.7 g
58	≈ 2 g
59	$0°C$; ice : water $= 1:5$
60	$10°C$ drop